A

Mini Project Report

On

QuickBites

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science and Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on Quick Bites has been submitted by Razzak Ansari (24107097), Vishal Chauhan (24107072), Mihir Deshmukh (24107049), Bhoomi Lokhande (24107078) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science and Engineering (Data Science), during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Ms. Richa Singh

Guide

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1.

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Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide Ms. Richa Singh. We express our gratitude towards our HOD, Dr. Pravin Adivarekar, and the Department of Computer Science and Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator Mr. Kiran Gaikwad and Ms.Richa Singh who gave us valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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CHAPTER 1

INTRODUCTION

In modern educational institutions, the college canteen acts as a central hub for students and faculty. However, many college canteens still rely on traditional manual methods for managing daily operations. These include physical queueing for order placement, cash-based transactions, and verbal communication for order readiness. Such systems are inefficient, especially during peak break hours, leading to long wait times and overall congestion.

The "QuickBites" project aims to provide a centralized digital platform that simplifies and automates food ordering and queue management. The application is designed to enhance the experience for students and staff by providing real-time menu access, live order updates, and secure user authentication, while giving canteen administrators a streamlined dashboard to process orders efficiently.

1.1 Purpose

The purpose of this document is to present the design, development, and implementation details of the Canteen Management System. It explains how modern backend technologies like Java, Spring Boot, and WebSockets can be utilized alongside responsive frontends (React/Flutter) to build a scalable and highly concurrent application tailored to an active college environment.

1.2 Problem Statement

During short academic breaks, the canteen experiences sudden, high-volume foot traffic. The manual system of taking orders and calling out token numbers creates bottlenecks. Students often spend the majority of their break waiting in lines rather than consuming their meals. Additionally, manual menu management makes it difficult for the canteen staff to update item availability in real-time, leading to order cancellations and frustration.

1.3 Objectives

1) To reduce rush and waiting time in college canteens by enabling students to place food orders in advance through a mobile.

2) To allow students to pay digitally, saving time and avoiding cash payments.

3) To improve order management for canteen staff by displaying real-time orders, reducing confusion and manual errors.

4) To enhance overall user convenience and efficiency by offering features like menu browsing, order tracking, and notifications when food is ready.

1.4 Scope

The scope covers the full lifecycle of a canteen order, from user registration and menu browsing to live queue tracking and admin order fulfillment. It is designed to be accessible via web and mobile interfaces. The current scope focuses on software-based queuing and order management and does not yet include physical hardware integrations like self-service kiosk machines.

CHAPTER 2

PROPOSED SYSTEM

The proposed system is a highly scalable application utilizing a robust Spring Boot architecture for the backend and React/Flutter for the frontend interfaces. It aims to digitize the canteen experience. Students and faculty can browse the digital menu, place orders, and track their preparation status in real-time. Canteen staff utilize a dedicated admin interface to update menus, manage stock, and advance order statuses.

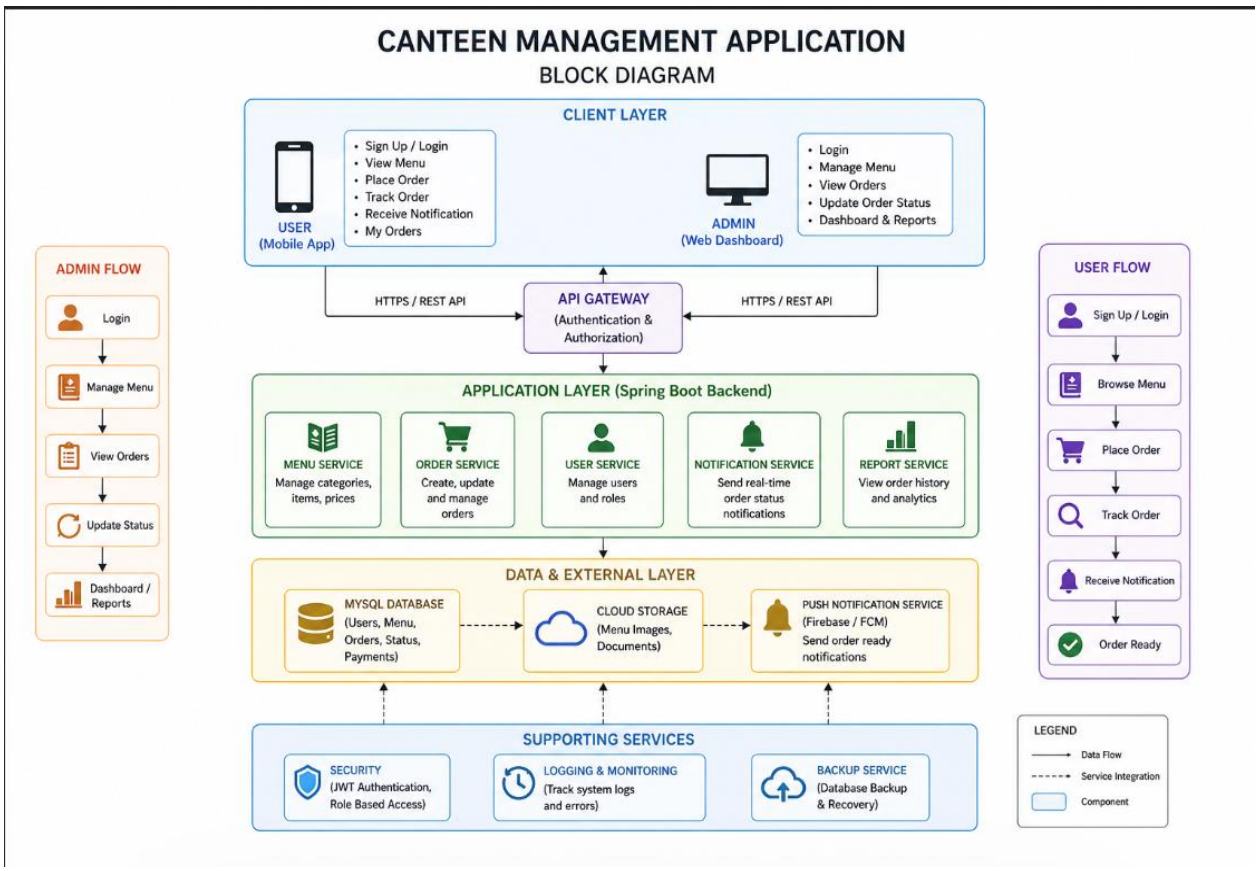


Fig 2.1 Block Diagram

Figure 2.1 block diagram represents the overall architecture of the system. The client layer includes user and admin interfaces. The API layer handles communication between frontend and backend. The application layer (Spring Boot) processes business logic, while the database stores user, menu, and order data. Notification services provide real-time updates to users.

2.1 Features and Functionality

User Features

- **Secure Authentication:** Students register and log in securely utilizing JWT authentication. This ensures safe access and protects user data.
- **Digital Menu Browsing:** Users can view available food items, prices, and high-quality images fetched from Cloudinary.
- **Live Order Tracking:** Utilizing WebSockets, users receive real-time updates on their order status (e.g., Pending, Preparing, Ready for Pickup) without needing to refresh the page.
- **Order History:** Users can access their past orders for quick re-ordering.

Admin Features

- **Menu Management:** Admins can seamlessly add, edit, or remove food items and upload images directly to Cloudinary.
- **Real-time Order Queue:** Admins view incoming orders instantly via WebSockets and can transition their states to manage the kitchen workflow.
- **Stock & Availability:** Admins can toggle the availability of items to prevent orders for out-of-stock food.

CHAPTER 3

PROJECT OUTCOMES

The implementation of the Quick Bites application has resulted in a modernized, efficient canteen environment. The primary outcome is the drastic reduction in physical queues. By allowing students to order from their classrooms or the campus grounds and arrive only when notified that their food is ready, the system optimizes time management for the student body.

Furthermore, the integration of WebSockets successfully eliminates communication delays between the kitchen and the customer. The use of JWT ensures that user sessions remain secure, while Cloudinary integration provides a fast, optimized delivery of menu visuals, keeping the core database lightweight. Data-Driven Management and Financial Transparency

Beyond the front-end user experience, the project provides a robust administrative outcome through real-time inventory and sales tracking. The app serves as a centralized hub for canteen managers to monitor which items are high-performing and which are stagnating, enabling smarter stock procurement and waste reduction. Furthermore, the integration of digital payment gateways and transaction history ensures complete financial transparency. This eliminates the common hassles of handling physical cash and providing "change," while giving the administration detailed analytical reports to evaluate the canteen's daily and monthly performance.

Enhanced User Satisfaction and Accessibility

From a user perspective, the project results in a modernized, user-centric interface that prioritizes convenience. Features such as live order status updates, digital receipts, and a dynamic menu (showing item availability) empower students to plan their breaks more effectively. This digital transformation fosters a more organized and professional environment, moving away from the chaotic nature of traditional canteen management. Ultimately, Quick Bites establishes a scalable digital infrastructure that can be easily updated with new features, such as feedback systems or loyalty rewards, ensuring the canteen evolves alongside the needs of the student body.

CHAPTER 4
SOFTWARE REQUIREMENTS

Frontend: Flutter (Mobile)

Used to build responsive, cross-platform user interfaces. They allow for dynamic state management to render live updates seamlessly.

Backend: Java & Spring Boot

Spring Boot provides the robust architectural framework necessary to handle high-volume API requests, dependency injection, and overall business logic.

Security: JWT (JSON Web Token)

Employed for stateless, secure user authentication and authorization across API endpoints.

Real-Time Communication: WebSockets

Facilitates full-duplex communication channels over a single TCP connection, powering the live order status updates.

Asset Management: Cloudinary

A cloud-based service utilized for hosting and optimizing menu item images, ensuring fast load times.

Database: MySQL

Used for structured data storage including user profiles, normalized order details, and menu items.

CHAPTER 5

PROJECT DESIGN

The application follows a decoupled Client-Server architecture. The React/Flutter frontends act as the client layer, consuming RESTful APIs exposed by the Spring Boot backend.

5.1 System Architecture

The architecture is organized into three primary layers: the Client Layer (user-facing apps), the Application Layer (Spring Boot backend with security, business logic, and WebSocket broker), and the Data & External Layer (MySQL database and Cloudinary CDN). The Security Gateway employing JWT sits at the boundary, intercepting all HTTP REST requests before they reach application services.

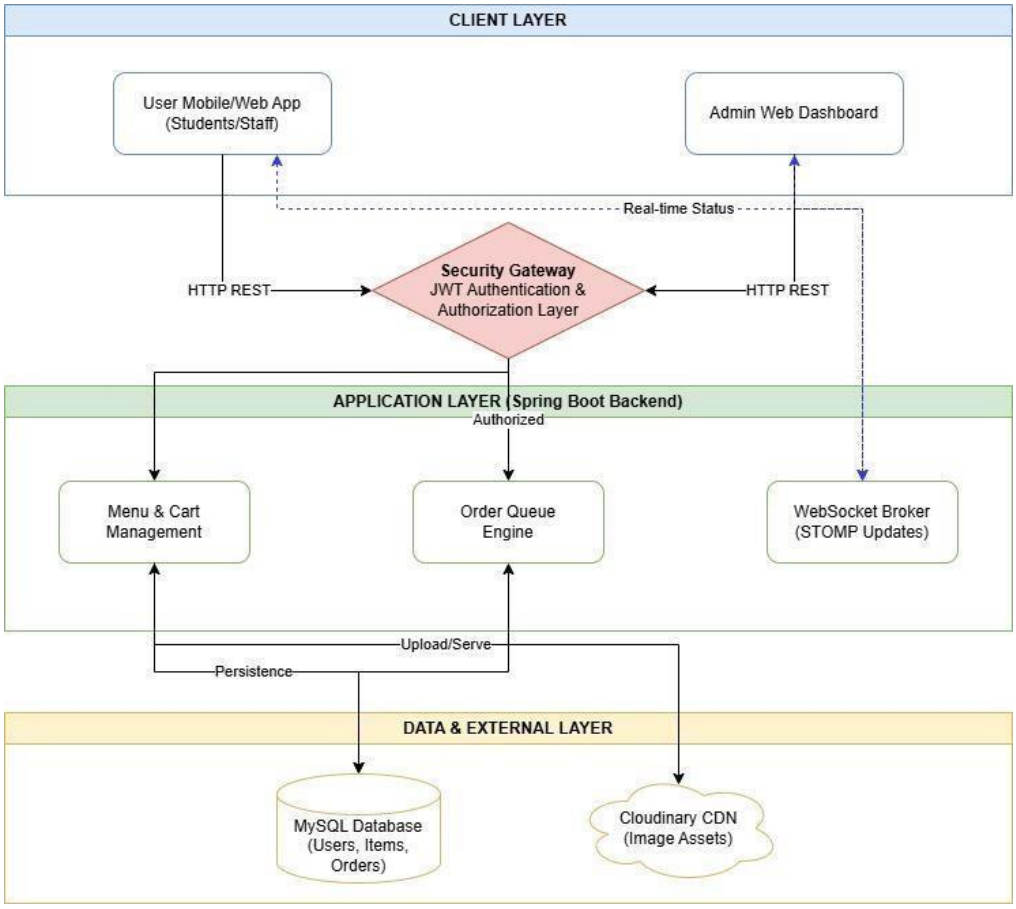


Fig 5.1: System Architecture – Client, Application, and Data Layers

5.2 System Workflow

1. The user authenticates via the login endpoint; the Spring Boot server validates credentials and returns a JWT.
2. The user browses the menu (images fetched from Cloudinary via URLs stored in the database).
3. Upon placing an order, a REST API call records the transaction in the database.
4. The WebSocket connection instantly broadcasts this new order to the Admin Dashboard.
5. As the admin updates the order status (e.g., to "Ready"), the WebSocket pushes the update directly back to the specific user's client.

5.3 Application Workflow

The following diagram illustrates the complete application flow for both Admin and User roles.

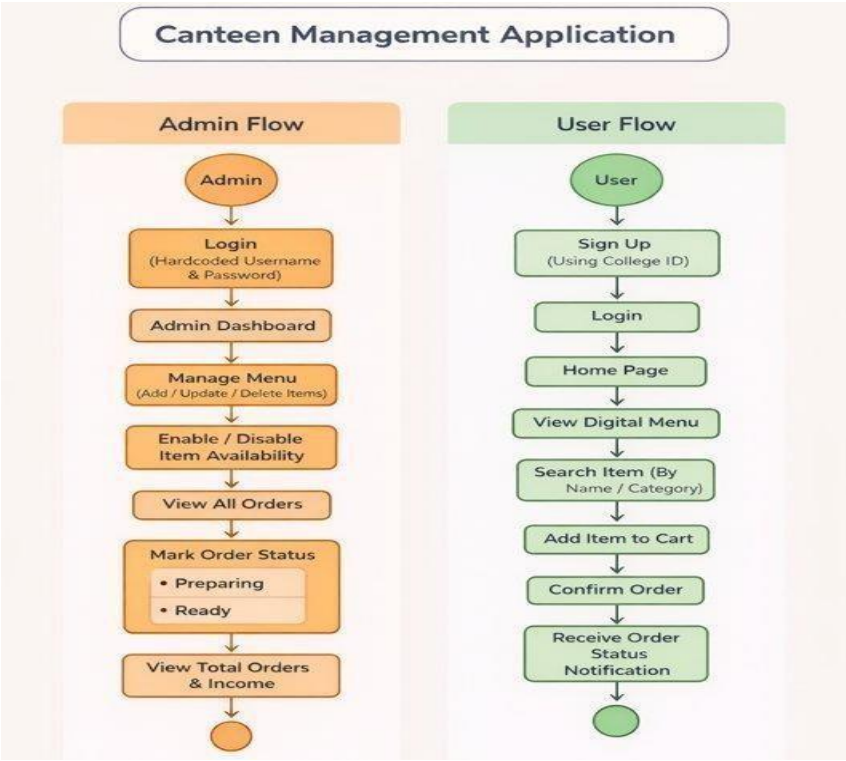


Fig 5.3: Application Workflow – Admin and User Flow

5.4 Security Design

Endpoints modifying data or accessing personal history are intercepted by security filters that validate the attached JWT. Role-based access ensures standard users cannot access the kitchen's active queue endpoints.

CHAPTER 6

PROJECT SCHEDULING

The project was executed over an academic timeline consisting of requirement gathering, system design, implementation, and testing phases.

The project was executed in multiple stages, starting with database design and system planning, followed by backend development using Spring Boot and frontend development using Flutter. Real-time features were then integrated to enable live updates and smooth communication between users and admin. Finally, the system was thoroughly tested and refined to ensure proper functionality, performance, and reliability.

GANTT CHART TEMPLATE

Smartsheet Tip • A Gantt chart's visual timeline allows you to see details about each task as well as project dependencies.

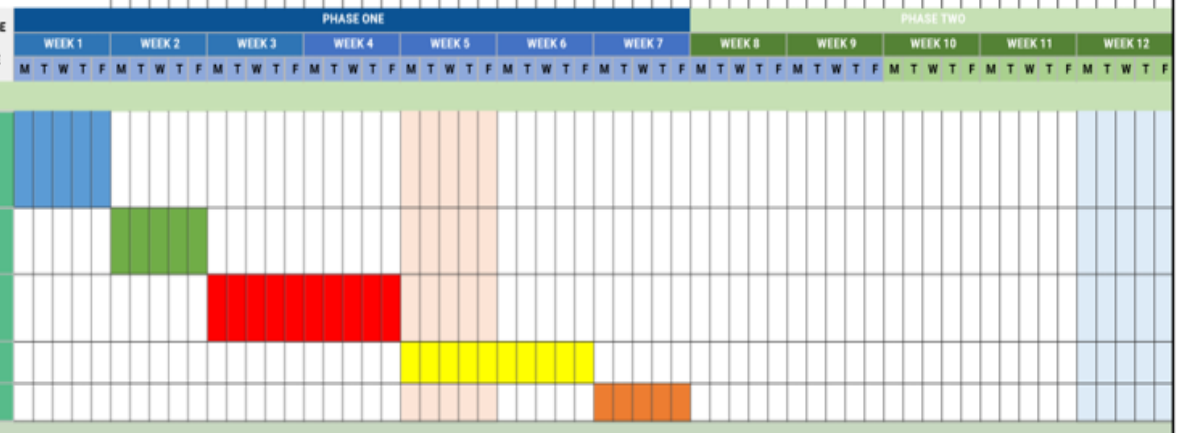
PROJECT TITLE: Quick Bites – Smart Canteen Management System

PROJECT GUIDE: Ms. Richa Singh

INSTUTUTE & DEPARTMENT NAME: AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

DATE: 4/1/26

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	Vishal	1/13/26	1/16/26	1	100%
1.2	Identifying the functionalities of the Mini Project	Razzak, Mihir, Bhoomi, Vishal	1/16/26	1/20/26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	Razzak, Mihir, Bhoomi, Vishal	1/20/26	2/3/26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Mihir	2/3/26	2/17/26	2	100%
1.5	Presentation I	Razzak, Mihir, Bhoomi, Vishal	2/17/26	2/24/26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	Razzak	2/24/26	3/10/26	2	100%
2.2	Database Connectivity of all modules	Razzak	3/10/26	3/17/26	1	100%
2.3	Integration of all modules and Report Writing	Bhoomi	3/17/26	3/24/26	1	100%
2.4	Presentation II	Razzak, Mihir, Bhoomi, Vishal	3/24/26	4/1/26	1	100%



CHAPTER 7: RESULTS

The system was tested under simulated peak-hour loads to ensure reliability. The following screenshots demonstrate the working application across key user journeys.

Home / Digital Menu Screen

Figure 7.1 shows the home screen of the APSIT Canteen Management mobile application. The screen displays a search bar at the top for finding dishes quickly, a promotional banner highlighting popular campus items, and category filters (All Items, Breakfast, etc.). The "Popular Today" section lists items like Aloo Paratha and Bhel Puri with their preparation times, categories, prices, and an "Add to Cart" button. The bottom navigation bar provides quick access to Home, Cart, Orders, and Profile sections.

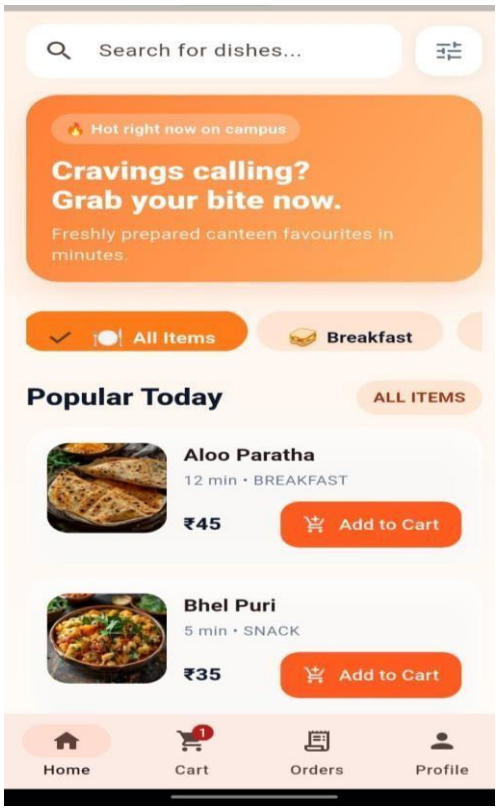


Fig 7.1: Home Screen – Digital Menu with Popular Items

Cart and Order Placement Screen

Figure 7.2 illustrates the cart screen where a student can review their selected items before placing an order. The screen shows an estimated preparation time of "About 1 min," the item details (Bhel Puri – SNACK, ₹35), quantity controls, and a summary showing the total amount inclusive of all taxes. A prominent "Place Order" button at the bottom allows the student to confirm their order.

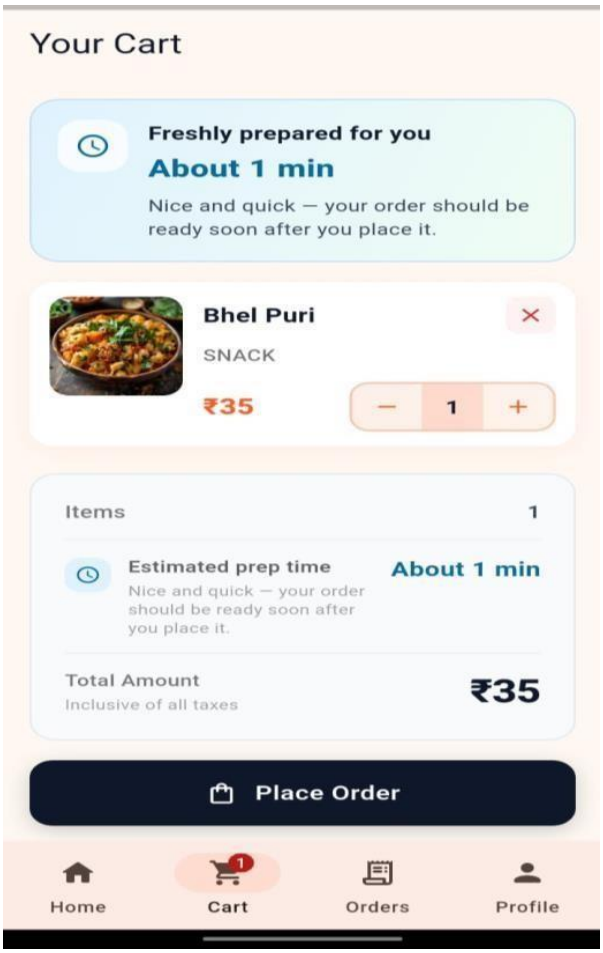


Fig 7.2: Cart Screen – Item Review and Order Placement

My Orders Screen with Live Queue Insight

Figure 7.3 shows the My Orders screen which displays the user's order history categorized by recency. At the top, a "Live Queue Insight" banner shows the current overall queue wait time. Each order card displays the total amount, placement timestamp, number of items, and current status (READY). This allows students to track their orders at a glance without visiting the counter.

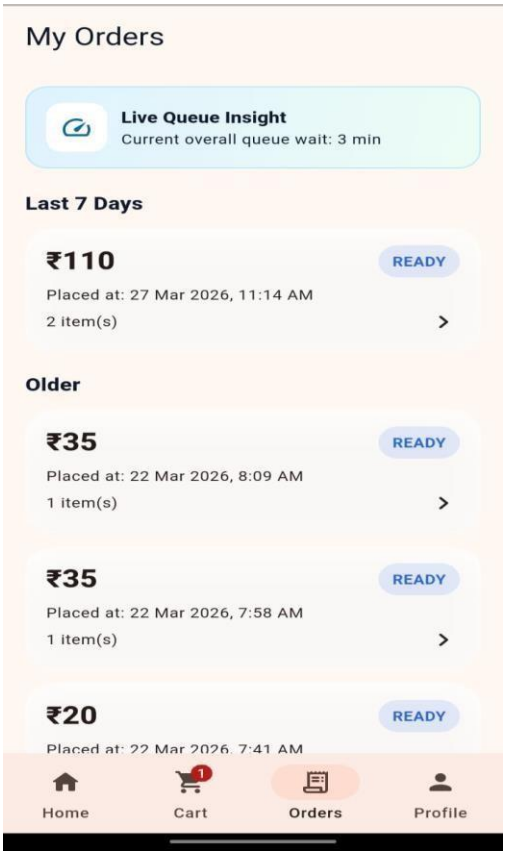


Fig 7.3: My Orders Screen – Live Queue Insight and Order History

Order Details with QR Code

Figure 7.4 demonstrates the order details screen which is displayed when an order status is marked as "READY" by the admin. A prominent "Your Order is Ready!" message appears along with a dynamically generated QR code that the student must show at the counter to claim their food. The order code (e.g., AD3E1253), order ID, placed timestamp, and itemized bill are also displayed, fully eliminating the need for physical token systems.

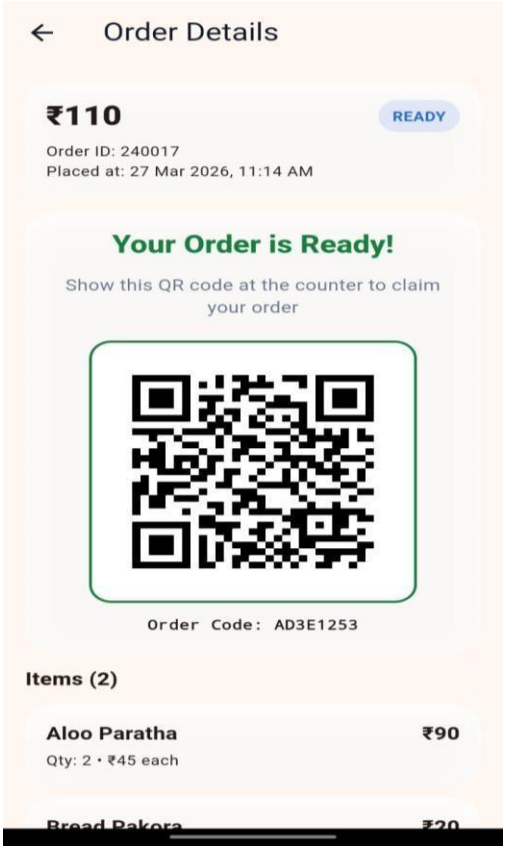


Fig 7.4: Order Details – QR Code for Contactless Order Pickup

Profile Screen

Figure 7.5 shows the user profile screen which displays the student's username, college email ID, and registered phone number. The screen offers a "Change Password" option for account security and a clearly visible "Logout" button. The consistent bottom navigation bar ensures seamless navigation throughout the application.

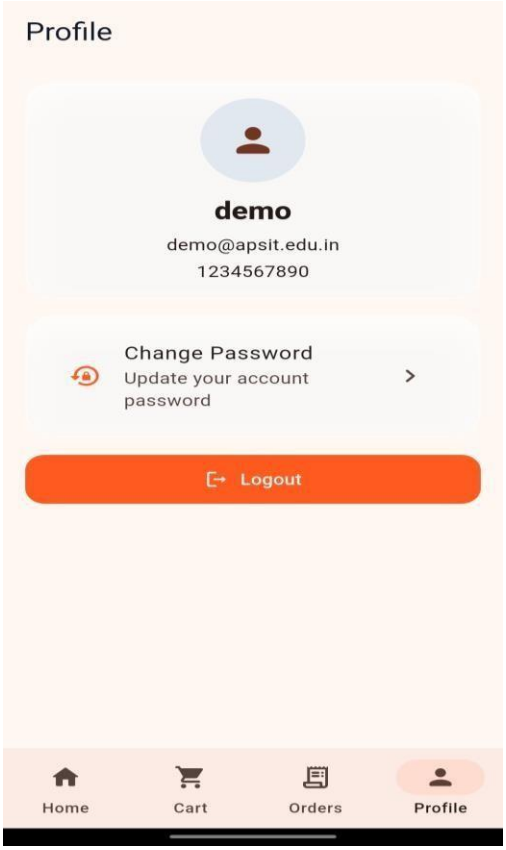


Fig 7.5: Profile Screen – User Account Management

CHAPTER 8

CONCLUSION

The QuickBites application successfully bridges the gap between traditional food service and modern digital convenience. By architecting a solution with Spring Boot, React/Flutter, and WebSockets, the project addresses the specific pain points of college break-time rushes. The system provides a highly responsive, secure, and user-friendly experience, demonstrating the practical application of full-stack development and real-time data handling to solve localized logistical challenges.

The QR code-based pickup system, live queue insights, and real-time status updates collectively eliminate the need for physical token systems and reduce counter congestion. The modular architecture ensures the system can be scaled and extended with features such as payment gateway integration, analytics dashboards, and IoT-based hardware in future iterations.

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